CHOOSING THE RIGHT PAAS

Platform as a Service (PaaS) is a cloud based environment that supports continuous development, continuous integration, and the management of applications. Since it can be a challenge to choose the right PaaS for your needs, here is a guide that will help you make a decision.

Cloud computing is a disruptive technology that delivers services on demand with a pay-as-you-go billing model. According to the definition of NIST (the US government's National Institute of Standards and Technology), there are four cloud deployment models—public clouds, private clouds, hybrid clouds and community clouds. And there are three cloud service models, namely, Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS).

PaaS is abstracted from the lower-level physical or virtual infrastructure components. In PaaS, the service provider is responsible for provisioning, de-provisioning and managing the infrastructure resources. It also offers fully managed application development and deployment services. PaaS provides developers with the flavours of operating systems, databases, middleware, software tools and managed services. In PaaS, the following features are available:

- Users can deploy applications onto Agile infrastructure using supported programming languages such as Java, PHP, Ruby, .NET
- Access to platforms/tools such as Web servers/application servers
- Databases as a service
- PaaS providers manage the underlying infrastructure resources
- They manage the Web servers, application servers, databases, operating systems, virtual servers, networks, back-up, disaster recovery, etc.

Thus PaaS users or organisations can focus on the most significant aspects of their businesses rather than worrying about managing resources, platforms and versions.

PaaS includes not only a deployment environment, but also repositories such as a build environment, a testing environment, performance management, mail services, log services, database services, Big Data services, search services, enterprise messaging services, application performance management for modern application architectures, code inspection services, etc.

Choosing the appropriate PaaS provider for your business can be quite a confusing task, because of the ever changing technology landscape and players in this market. The major factor to evaluate right at the beginning whether you want to select a PaaS provider or platform that supports the most number of enterprise applications, with diverse technologies. Apart from that, the PaaS platform should be easy to operate.

Needless to say, all organisations would want to invest in a PaaS platform that delivers performance, agility and scalability along with adequate coverage, and offers long term benefits. So, before you get on with the job of finalising a PaaS platform for your organisation, do go through our detailed buyers' guide.

Business drivers for using PaaS

PaaS is driving an era of innovation, agility, and faster time to market. Listed below are some of the vital business drivers:

- PaaS enablement is faster as a runtime environment is available out-of-the-box
- Access to all application resources is based on the type of PaaS—public or private
- Upgradation of the middle layer is managed by the service provider and the PaaS consumer is not responsible for it
- Scope for innovation is higher, as the major focus remains on applications and not on maintaining and managing infrastructure
- Higher cost benefit with transparency
- Rapid and easy scalability

Common questions to help you evaluate a PaaS platform

Here are a few questions that you need to answer in order to choose the best solution for you or your organisation.

- **How does PaaS fit into the overall cloud strategy of your organisation?**

In the current scenario, DevOps and PaaS are more related, and both can have a significant impact only if they suit the culture of an organisation. How useful DevOps or PaaS is within a firm's existing culture is a prime factor for successful PaaS enablement.

- **What are the languages, runtime, databases, etc, supported by the PaaS provider?**

It is important to select a PaaS offering that supports multiple programming languages, databases, NoSQL databases, messaging services, etc.

Other aspects that should also be looked into are:

- Who are the developers of the PaaS platform?
- In which language is the platform written and what type of licence does it have?
- Is the service provider offering the most striking benefits of using PaaS to your organisation?
- What are the top options for Java, .NET, and the other primary languages of the organisation? Is there any good open source alternative available in the market?